

## 1.2 R Packages and the Tidyverse

Installing packages, loading them, the tidyverse, and the pipe.

### 1. What Is a Package?

R already contains many functions:

```
mean()  
sum()  
sqrt()  
seq()  
data.frame()
```

These functions come with **Base R** and the standard R installation.

However, not every useful function is included in Base R.

R can be extended using **packages**.

A **package** is a collection of functions, data sets, documentation, and other tools designed for particular tasks.

Thousands of R packages are available for statistics, visualization, machine learning, data manipulation, and many other applications.

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### 2. Installing a Package

Before using a package for the first time, we usually need to **install** it.

For example:

```
install.packages("ggplot2")
```

The package name must be inside quotation marks:

```
install.packages("ggplot2")
```

not:

```
install.packages(ggplot2)
```

Installation downloads the package and places it on your computer.

## Usually, you install a package only once.

Think of installing an R package like installing an app on your computer.

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### 3. Loading a Package

Installing a package does **not** automatically make its functions available in the current R session.

We load an installed package using:

```
library(ggplot2)
```

Notice the difference:

```
install.packages("ggplot2")  # Install  
library(ggplot2)             # Load
```

A useful rule to remember is:

**Install once; load each new R session.**

For example, you may install `ggplot2` once:

```
install.packages("ggplot2")
```

Then, whenever you start a new R session and want to use it:

```
library(ggplot2)
```

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### 4. What Is an R Session?

An **R session** is the period during which R is currently running.

When you close and restart R or RStudio, you begin a new session.

Packages that you loaded previously are generally not automatically loaded into the new session.

Therefore, your script might begin with:

```
library(ggplot2)  
library(dplyr)
```

but you should **not normally put** this in a script that runs every time:

```
install.packages("ggplot2")
```

Installation is something you do when needed, not every time the script runs.

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## 5. Using a Function from a Package

Suppose we load `ggplot2`:

```
library(ggplot2)
```

We can now use functions provided by that package:

```
ggplot()
```

Another way to use a function is:

```
ggplot2::ggplot()
```

The notation:

```
package::function()
```

means:

"Use this function specifically from this package."

For example:

```
dplyr::filter()
```

means:

Use the `filter()` function from the `dplyr` package.

This can be useful because different packages can sometimes contain functions with the same name.

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## 6. Getting Help

R has a built-in help system.

To read the documentation for a function:

```
?mean
```

or:

```
help(mean)
```

For a function from a package:

```
?ggplot2::ggplot
```

You can also search the help system:

```
help.search("mean")
```

A shortcut is:

```
??mean
```

Therefore:

```
?function      → documentation for a function  
??topic        → search R help for a topic
```

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## 7. What Packages Are Currently Loaded?

You can see information about the current R session using:

```
sessionInfo()
```

You can also see the current search path:

```
search()
```

For example, after:

```
library(ggplot2)
```

try:

```
search()
```

and notice that `ggplot2` appears in the search path.

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## 8. What Packages Are Installed?

To see installed packages:

```
installed.packages()
```

This returns considerable information.

To see just their names:

```
rownames(installed.packages())
```

To check whether a particular package is available:

```
requireNamespace("ggplot2", quietly = TRUE)
```

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## 9. What Is the Tidyverse?

In this course, we will learn both **Base R** and the **tidyverse**.

The tidyverse is a collection of R packages designed to work together for data science.

We can install it using:

```
install.packages("tidyverse")
```

and load it using:

```
library(tidyverse)
```

Loading `tidyverse` loads several important packages.

Some of the most commonly used are:

| Package              | Main purpose                          |
|----------------------|---------------------------------------|
| <code>ggplot2</code> | Data visualization                    |
| <code>dplyr</code>   | Data manipulation                     |
| <code>tidyr</code>   | Reshaping and tidying data            |
| <code>readr</code>   | Reading data files                    |
| <code>tibble</code>  | Modern data frames                    |
| <code>stringr</code> | Working with strings                  |
| <code>forcats</code> | Working with categorical data/factors |
| <code>purrr</code>   | Functional programming and iteration  |

We will encounter these packages gradually throughout the course.

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## 10. Base R vs. Tidyverse

Often, the same task can be performed using either Base R or tidyverse tools.

For example, suppose:

```
students <- data.frame(  
  name = c("John", "Sara", "David"),  
  grade = c(75, 92, 85)  
)
```

Using **Base R**, we could select students with grades above 80:

```
students[students$grade > 80, ]
```

or, with `subset()`:

```
subset(students, grade > 80)
```

`subset()` lets you write the condition with the column name. In dplyr, two functions sit next to it:

- `filter()` is the closest match for keeping rows
- `select()` is the closest match for keeping columns (`subset(students, select = name)`)

```
library(dplyr)  
  
filter(students, grade > 80)
```

Both the Base R and dplyr approaches are valid.

Throughout this course, we will learn how to work with data using both approaches.

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## 11. The Pipe

One important idea in modern R and the tidyverse is the **pipe**.

Suppose:

```
x <- c(10, 20, 30, 40)
```

We could calculate its mean using:

```
mean(x)
```

Base R provides the pipe:

```
x |> mean()
```

You can read this approximately as:

Take `x`, **then** calculate its mean.

The tidyverse historically popularized another pipe:

```
x %>% mean()
```

The `%>%` pipe is provided by `magrittr` and is commonly encountered in tidyverse code.

In this course, you may encounter both:

```
|>      # Base R pipe

%>%     # traditional tidyverse pipe
```

The pipe becomes especially useful when several operations are performed in sequence.

For example:

```
students |>
  subset(grade > 80) |>
  head()
```

`subset()` is Base R. The dplyr line that does the same row-keeping is `filter(grade > 80)`. Note 2.1 goes through `subset()`, `filter()`, and `select()` more carefully.

Later, we will use pipes extensively with tidyverse functions.

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## 12. A Typical R Workflow

A simple R project might follow this pattern:

### Step 1 — Install required packages

Usually done only once:

```
install.packages("tidyverse")
```

### Step 2 — Load packages

Done when beginning a new R session:

```
library(tidyverse)
```

### Step 3 — Import or create data

We can create a small data set in code:

```
students <- data.frame(  
  name = c("John", "Sara", "David"),  
  grade = c(75, 92, 85)  
)
```

We can also read a file. Download [students.csv](#) and place it in your working directory. Then:

```
students <- read.csv("students.csv")
```

With the tidyverse, an equivalent approach is:

```
students <- readr::read_csv("students.csv")
```

`read.csv()` is Base R. `read_csv()` comes from the `readr` package.

Note **1.3** is the full treatment: relative paths, delimiters, `read.table()` / `read_csv()`, inspecting the import, and exporting.

### Step 4 — Explore the data

```
head(students)  
str(students)  
summary(students)  
View(students)
```

### Step 5 — Manipulate the data

```
students |>  
  dplyr::filter(grade > 80)
```

### Step 6 — Analyze or visualize the data

For example:

```
mean(students$grade)
```

or later:

```
ggplot(students, aes(x = name, y = grade)) +  
  geom_col()
```

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## 13. Important Package Commands to Remember

```
install.packages("package")
```

**Install** a package.

```
library(package)
```

**Load** an installed package.

```
package::function()
```

Use a particular **function from a particular package**.

```
?function
```

Read a function's **documentation**.

```
??topic
```

**Search** R's help system.

```
installed.packages()
```

See the packages that are **installed**.

```
sessionInfo()
```

See information about the current **R session**.

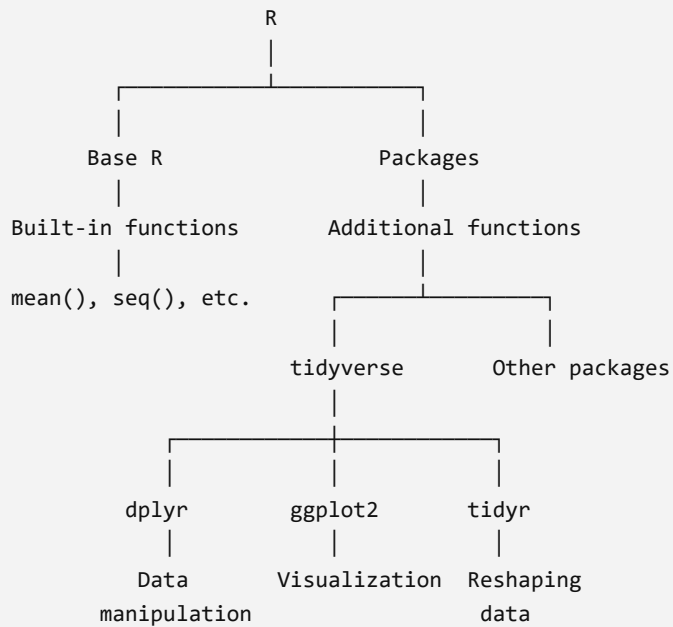
```
search()
```

See packages currently attached to the **search path**.

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## 14. The Big Picture

A useful way to think about the R ecosystem is:



The central distinction to remember is:

**R is the programming language. A package extends R by providing additional functions and tools.**

And:

**Base R and tidyverse are not two different programming languages. The tidyverse is a collection of packages that we use within R.**

For packages, remember the three most important commands:

```
install.packages("tidyverse") # Get it
library(tidyverse)           # Load it
dplyr::filter()              # Use a specific function
```